Regression modelling and optimisation of plasma parameters for controlled photon emission in plasma-enhanced atomic layer deposition

1. Abstract

Plasma-enhanced atomic layer deposition (PE-ALD) is a delicate process, relying on a balance of reactive radicals, ions, and photons. Research shows that vacuum ultraviolet (VUV) photon emissions are known to cause substrate damage even under benign ion conditions (Profijt et al., 2011b; Arts et al., 2022). This project investigates whether regression models are suitable for predicting and optimising oxygen plasma photon flux as a function of power and pressure, enabling rapid separation of VUV and non-VUV spectral components. Photon emission data were generated from a zero-dimensional plasma-chemical kinetics model (Engelbrecht et al., 2024) across 100–2000 W and 2–200 mTorr. One-dimensional trends were captured by an extended saturation function (training $R^2 > 0.99$), while a log-space interaction model reproduced the two-dimensional parameter space with a worst-case interpolation error below 10 %. The fitted surfaces reveal that tuning pressure and power in tandem can form a maximum VUV to non-VUV ratio of 1.51 (200 mTorr, 100 W) to a minimum of 0.36 (2 mTorr, 2000 W) within the parameter ranges, with more extreme ratios probable at more extreme parameter ranges. These results demonstrate that data-driven regression models can serve as computationally cheap and reasonable accurate ways of analysing plasma emission, offering a practical route to accelerated optimisation of plasma outputs with increasing numbers of changing plasma parameters; a great help for the mitigation of VUV-induced damage in PE-ALD.

2. Introduction

Plasma-Enhanced Atomic Layer Deposition (PE-ALD) is a cyclic thin-film growth technique that extends conventional ALD by using plasma-generated reactive species instead of, or in addition to thermal reactants. Each PE-ALD cycle typically consists of two half cycles, both of which ideally being self-limiting:

1. A chemical precursor adsorbs onto available surface sites, followed by a purge of excess precursor and byproducts.
2. The surface interacts with plasma-generated radicals, ions, and photons to complete the desired surface reaction and regenerate reactive sites, followed by a final purge of remaining reactive species, leaving a controlled monolayer deposition.

Compared to thermal ALD, PE-ALD allows lower deposition temperatures, enhanced reactivity, and greater control over film properties, including density, stoichiometry, and crystallinity (Profijt et al., 2011b; Arts et al., 2022).

PE-ALD introduces three main species produced from the plasma that interact with the surface during the second half cycle of the process, these being radicals, ions and photons. Controlling the flux and energy of these species is critical in performing successful PE-ALD, as these species both drive the surface reactions during the process, and can cause defects. (Arts et al., 2022)

Specifically, this project will investigate how the photon flux produced by oxygen plasmas changes with differing plasma powers and pressure (both singularly and in tandem) utilising the implementation of regression algorithms. It will then determined whether different wavelengths of photons (specifically VUV (Vacuum ultraviolet) photons vs non-VUV photons) are affected differently in order to separate desired wavelengths from undesired wavelengths for the application. The project will then aim to determine whether regression algorithms are a viable method for modelling plasma behaviour, and whether the implementation of such models could accelerate the finding of optimal plasma parameters for the application. The motivation for this stems from research indicating that VUV photons in the ~100nm range are one of the leading causes of degradation in certain cases of PE-ALD, specifically the charge carrier lifetime of processed silicon. (Profijt et al., 2011a) and therefore showing the importance of controlling the emission of these potentially harmful photon wavelengths.

3. Literature review

3.1 Effects of VUV photons in PE-ALD

In plasma-enhanced atomic layer deposition (PE-ALD), vacuum ultraviolet (VUV) photons form an additional energy-transfer mechanism between the plasma and the substrate. The review by Arts et al., 2022 notes that ultraviolet photons can enhance surface reactions, as in photon-assisted ALD, but can also damage the surfaces of sensitive electronic devices, meaning that photons must be considered as both a potentially useful and potentially harmful plasma species. It also states that plasma composition can be adjusted by changing the input power and gas pressure, which provides a route to controlling the photon environment at the substrate.

This balance is particularly important for VUV radiation. The review by Profijt et al., 2011 showed that VUV photons with energies up to 9.5 eV present in an oxygen plasma can degrade the electrical performance of electronic structures containing ALD-grown metal oxide films, even when ion energies are low enough to avoid substantial ion-induced damage. This indicates that photon-driven effects must be treated separately from ion bombardment in PE-ALD, because VUV radiation can alter film quality and device performance through direct interaction with the growing surface.

Consequently, VUV photons should be regarded as a controllable process parameter rather than an unavoidable by-product of the plasma. Their influence depends on both their absolute flux and their relative contribution within the plasma emission spectrum, which makes the VUV to non-VUV photon ratio a relevant quantity when considering plasma optimisation for PE-ALD.

3.2 Global plasma model for Ar/O_2 ICPs (Engelbrecht et al., 2024)

The simulation data used in this work are generated from a zero-dimensional, volume-averaged plasma-chemical kinetics model described by Engelbrecht et al. (2024). The model treats the inductively coupled plasma as spatially uniform, solving coupled particle balance and electron energy balance equations to obtain self-consistent species densities and electron temperature.

For each species s, the particle balance is

$$\frac{dn_s}{dt} = \sum_r \delta_{s,r} K_r \prod_j n_{r,j} - \frac{A}{V}\Gamma_{s,\text{wall}}, \qquad 3.2.1$$

where K_r are reaction rate coefficients (electron-impact rates evaluated as functions of electron temperature), $\delta_{s,r}$ are stoichiometric coefficients, and the term $(A/V)\Gamma_{s,\text{wall}}$ accounts for losses to the reactor walls. The surface-to-volume ratio A/V is calculated for a cylindrical geometry, directly coupling reactor dimensions to the plasma chemistry.

The electron temperature is obtained from the power balance

$$\frac{3}{2}k_B\frac{d(T_e n_e)}{dt} = S_{\text{abs}} + S_{\text{elast}} + S_{\text{inelast}} + S_{\text{flux}}, \qquad 3.2.2$$

with the absorbed power density $S_{\text{abs}} = P_{\text{in}}/V$. Wall fluxes for positive ions are computed via the Bohm velocity $u_B = \sqrt{k_B T_e/m_i}$ and sheath-edge density factors that relate the central density to the sheath edge.

The model incorporates an extensive reaction set for Ar/O_2 plasmas, including electrons, argon and oxygen neutrals, ions, metastables, and excited states. Radiative emission is calculated from the populations of emitting states. For a transition $a \rightarrow b$ the volumetric emission rate is

$$I_{ab} = \gamma_{ab} A_{ab} n_a, \qquad 3.2.3$$

where A_{ab} is the spontaneous emission coefficient and γ_{ab} is an escape factor accounting for self-absorption, modelled as a function of the line-centre absorption coefficient and reactor radius (Engelbrecht et al., 2024).

For the VUV region, the dominant features are the 130 nm and 135 nm oxygen lines arising from the O(3s ^{3}S) and O(3s ^{5}S) states, respectively. The 130 nm line is primarily produced via a multi-step pathway: electron-impact excitation first populates O(3P) states, which then radiatively decay into O(3S) and subsequently to the ground state. The model captures these cascading processes and shows that VUV emission intensity increases with power and exhibits a maximum at intermediate pressures, where electron density and atomic oxygen production are optimised before collisional quenching becomes dominant. This pressure-dependent optimum and the power-saturation behaviour are the physical signatures that the regression models in this project aim to capture.

4. Method

4.1 Plasma simulation model

The plasma emission data used in this work were generated by a simulation code that is not publicly available, but whose governing equations are described in detail by Engelbrecht et al. (2024) and summarised in Section 3.2. The model implements a self-consistent, zero-dimensional, volume-averaged plasma-chemical kinetics framework for Ar/O_2 inductively coupled plasmas, solving coupled particle and power balances to obtain species densities, electron temperature, and radiative emission rates. Reactor dimensions were taken from the ICP tube described by Kessels et al. (2007): diameter 65 mm and estimated plasma length 100 mm, giving a surface-to-volume ratio $A/V \approx 120\ \text{m}^{-1}$. Simulations were performed over a

power range of 100–2000 W and a pressure range of 0.3–100 Pa (2–750 mTorr). One-dimensional sweeps fixed pressure at 10 mTorr or power at 500 W; a two-dimensional grid of 5 powers × 10 pressures was used for the combined parameter study. The model output provides photon emission rates for 12 oxygen lines across VUV and non-VUV wavelengths, which served as the dataset for the regression analysis.

4.2 Obtaining results

The plasma simulation model used for this project produced 12 oxygen emission lines that will be used for the later data analysis, these emission lines can be found in the figure 4.2.1. The methods for implementing these 12 oxygen emission lines by the plasma model can be found in Engelbrecht et al., 2024.

Oxygen emission line	Wavelength (nm)
(VUV)	-
O_5s -> O;	135.6
O_5s -> O;	135.9
O_3s -> O;	130.2
O_3s -> O;	130.5
O_3s -> O;	130.6
(non-VUV)	-
O_1s -> O_1d	557.7
O_5p -> O_5s	777.2
O_5p -> O_5s	777.4
O_5p -> O_5s	777.5
O_3p -> O_3s	844.6
O_3p -> O_3s	844.6
O_3p -> O_3s	844.7

Fig 4.2.1. Table showing the oxygen emission lines produced and their respective wavelengths

The desired results for this project will be obtained by running plasma simulations on the above described model with differing powers and pressures. This will be done with one dimensional varying power and pressure data fixed at a set pressure and power respectively, as well as two dimensional varying power and pressure data in tandem.

The resulting photon flux data will then be used to train regression models of multiple types utilising SciPy's curve fitting function (SciPy, 2019), utilising a training and validation portion of the data to ensure model accuracy. The validation portion of the data will be taken from the latter half of each parameter range rather than as a percentage of overall data points, this is done to better analyse how the model interpolates across an expanded parameter range. In the case of the two dimensional power and pressure regression data, a full validation model will not be utilised due to the larger amount of datapoints that would be required to do so and still ensure sufficient data is provided for the model training, as such large amounts of data are outside the scope of this project. Instead, a single point interpolation validation test will be performed as this should be a sufficient accuracy for the more qualitative results that will be drawn from this data.

The resulting R^2 values from the regression fits as well as the produced figures will be used to justify the suitability of each model. From this, a final regression model will be determined for the power data, pressure data and two dimensional power and pressure data to be used in further parts of this project.

The finalised regression model will then be used quantitively analyse whether tuning both plasma pressure and power is a valid method of controlling specific photon emission wavelengths (specifically VUV versus non-VUV) in plasma reactors. This will be done by using the regression models to find the power and / or pressure values that the following peaks occur at:

- Peak total photon flux
- Peak VUV photon flux
- Peak non-VUV photon flux
- Peak VUV to non-VUV photon flux ratio
- Peak non-VUV to VUV photon flux ratio

Additionally, the regression results will be used to qualitatively determine whether using regression algorithms to model plasma outputs is an appropriate method for accelerating plasma parameter tuning, and whether these methods could be incorporated into increasing numbers of independent plasma control parameters, where finding ideal parameters for the plasma use case could prove time consuming if performed through trial and error.

5. Results

5.1 Part A (Separated power and pressure)

5.1.1 Initial data acquisition

Initial data collected from the plasma simulation model showed how photon flux varied with changing power and pressure. When power was increased between the range 100 and 2000W, all photon emission lines experienced near linear increases in flux, with slight damping occurring towards the upper end of the data range, as can be seen in figure 5.1.1.1. When pressure was increased between the range 2 and 750mTorr, all photon emission lines experienced a peak in flux towards the lower end of the data range, as can be seen in figure 5.1.1.2. This occurred between 10 and 20 mTorr for all emission lines, except notably the $O_1s > O_1d$ emission line, where the peak occurred around 40 mTorr, as seen in figure 5.1.1.3.

A higher resolution of pressure data points between 2 – 50mTorr due to this being the main area of significance in photon fluxes with changing pressures.

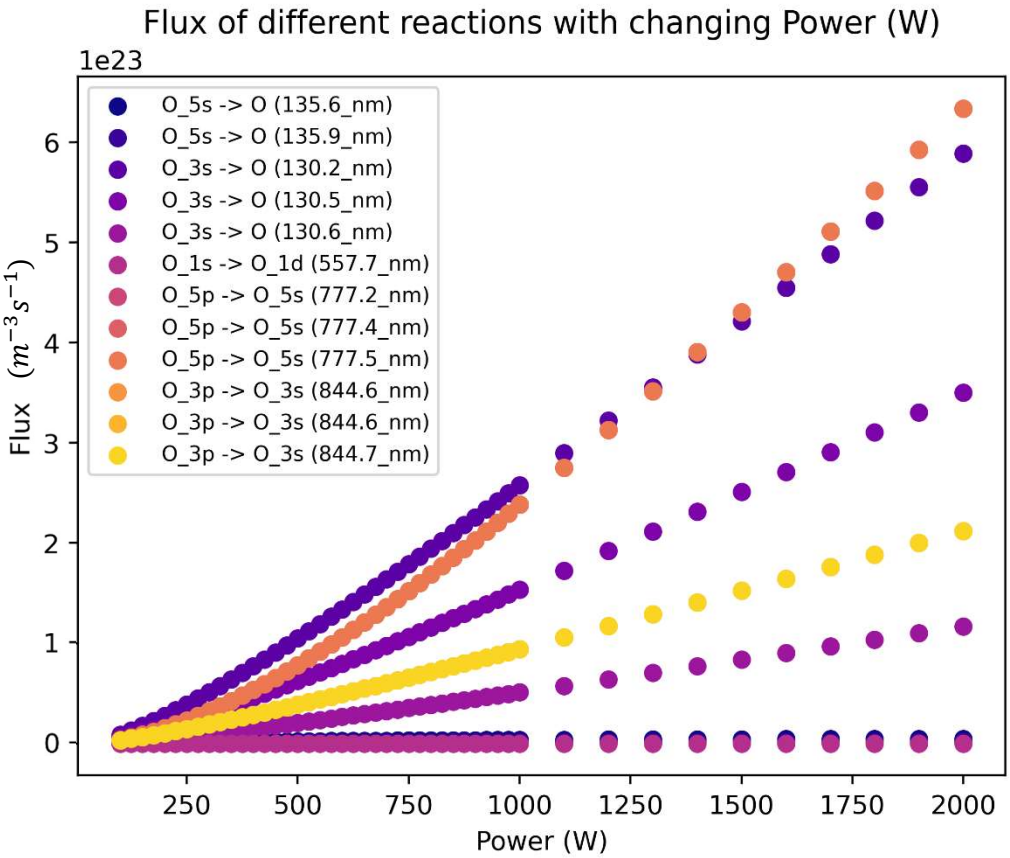


Fig 5.1.1.1. Graph showing the flux of different reactions with changing power

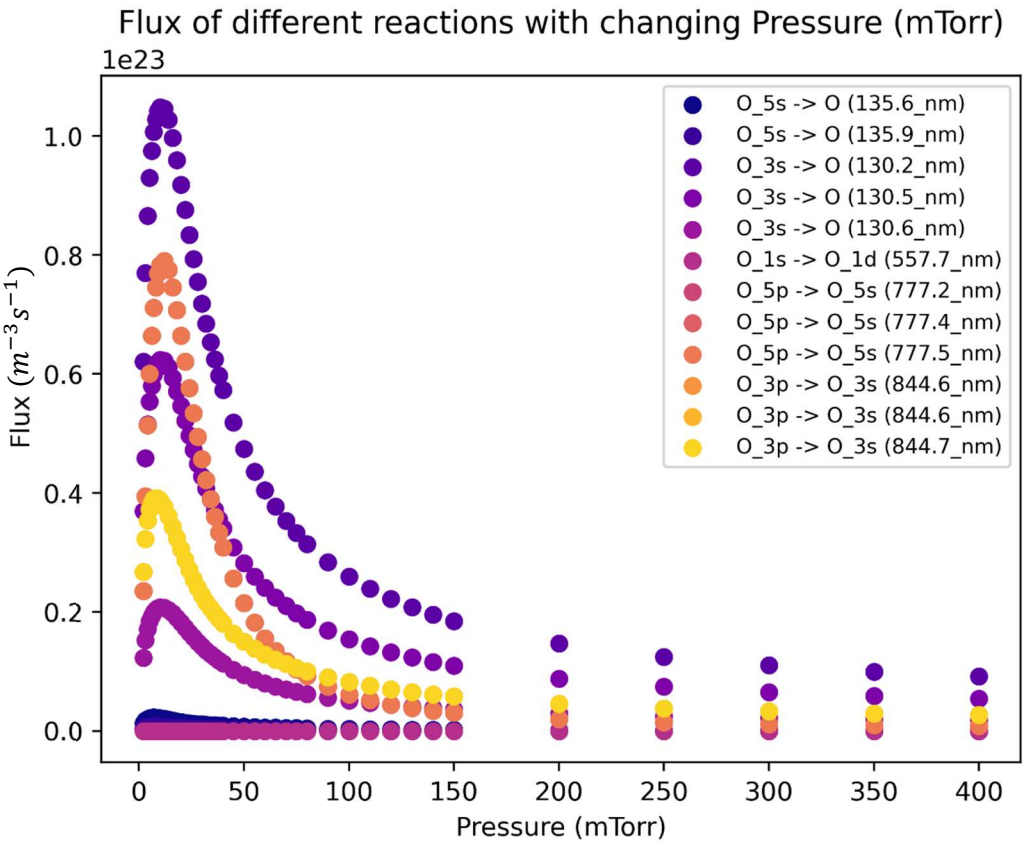


Fig 5.1.1.2. Graph showing the flux of different reactions with changing pressure

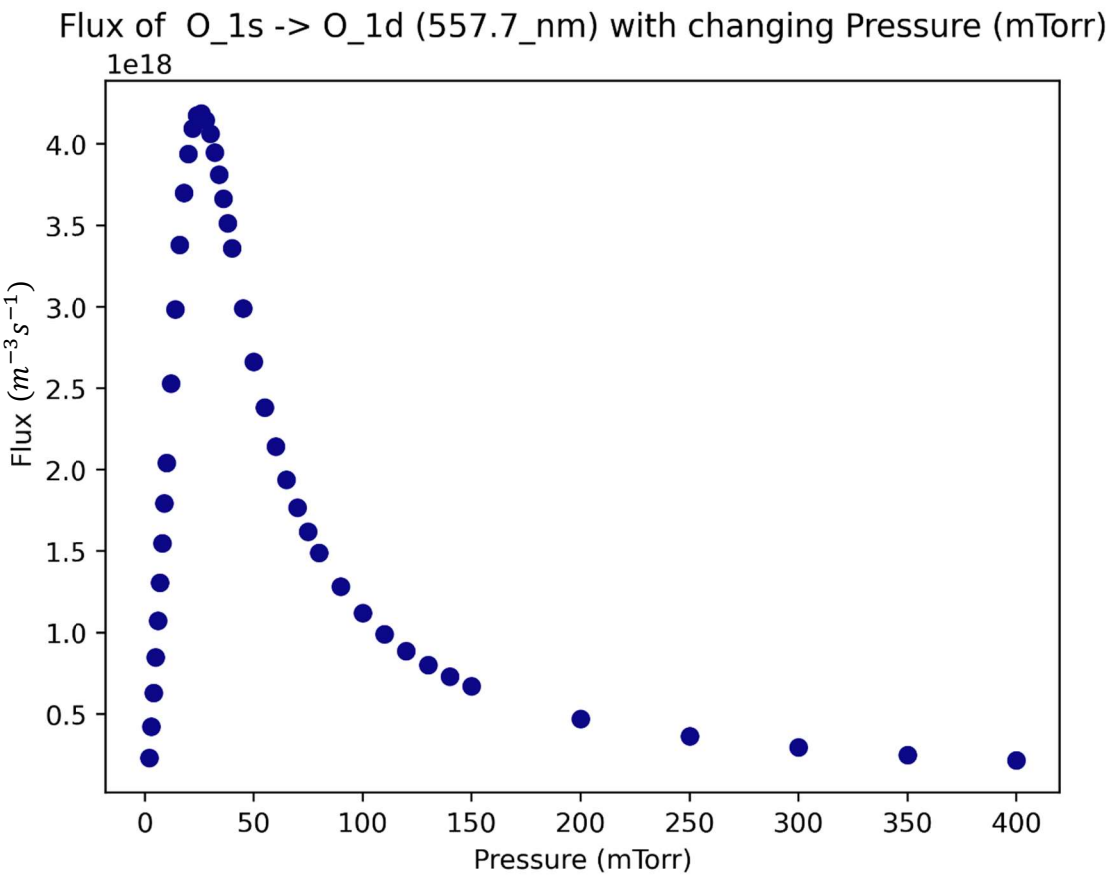


Fig 5.1.1.2. Graph showing the flux of the $O_1s > O_1d$ line with changing pressure

5.1.2 Regression model implementation

The following regression models were implemented to analyse both sets of data:

- Linear model: $I = \mathrm{m}x + c$
- Power law model: $I = A\mathrm{x}^{\alpha}$
- Saturation model: $I = \frac{A\mathrm{x}^{\alpha}}{1+B\mathrm{x}}$
- Extended saturation model: $I = \frac{A\mathrm{x}^{\alpha}}{1+B\mathrm{x}+C\mathrm{x}^{2}}$
- Exponential suppression model: $I = A\mathrm{x}^{\alpha}e^{-\gamma \mathrm{x}}$

The R^2 values for each model's training and validation data were analysed for each oxygen emission line to determine the model suitability.

The extended saturation model was determined as the most suitable regression model for measuring both the power data and the pressure data due to it having the highest R^2 for both data sets, as can be seen in figures 5.1.2.1 and 5.1.2.2, both which show the lowest R^2 value for each model across all emission lines. It is to note that the linear and power law model's do not have regression results for the pressure data due to them not being able to model the photon flux peak present in said data.

Model	Minimum regression R^2	Minimum validation R^2
Linear	0.981613	N/A
Power law	0.971568	N/A

Saturation	0.998422	0.950009
Extended saturation	0.999995	0.984202
Exponential suppression	0.999856	0.970786

Fig 5.1.2.1. Table showing the R^2 values for different regression models with changing power, both initial and validation

Model	Minimum regression R^2	Minimum validation R^2
Saturation	0.903586	0.927114
Extended saturation	0.995535	0.996308
Exponential suppression	0.944661	0.952461

Fig 5.1.2.2. Table showing the R^2 values for different regression models with changing pressure, both initial and validation

5.1.3 Regression model analysis

These models were then used to investigate how the total VUV and non-VUV flux changes with both changing pressures and powers, including analysing how the ratio between VUV and non-VUV flux changes, where peak total emission occurs, and for the case of changing pressure, at what values peak VUV and non-VUV emissions occur. It is to note that the pressure range was decreased to 2 – 200mTorr for this portion of the project, due to this encapsulation the main portions of interest of the data.

As shown in figure 5.1.3.1, the total flux peaks at 2000W, however it can be assumed from the established trends that the total flux value would continue to increase with increasing power. The maximum VUV to non-VUV ratio of 0.77 occurs at 100W and the minimum VUV to non-VUV ratio of 0.42 occurs at 2000W, with the ratio varying in a reciprocal manner across the data range. From this it can be assumed that the ratio would increase further at lower powers and decrease further at higher powers.

As shown in figure 5.1.3.2, the total flux peaks at 10.6 mTorr, with the VUV and non-VUV flux peaking at similar pressure values, being 9.3 and 11.0 mTorr respectively. The maximum VUV to non-VUV ratio of 1.14 occurs at 200mTorr and can once again be assumed to increase past this value for higher pressures. The minimum VUV to non-VUV ratio of 0.53 occurs at 13.4 mTorr, with the VUV to non-VUV ratio varying across the pressure range in a manner similar to the total flux flipped.

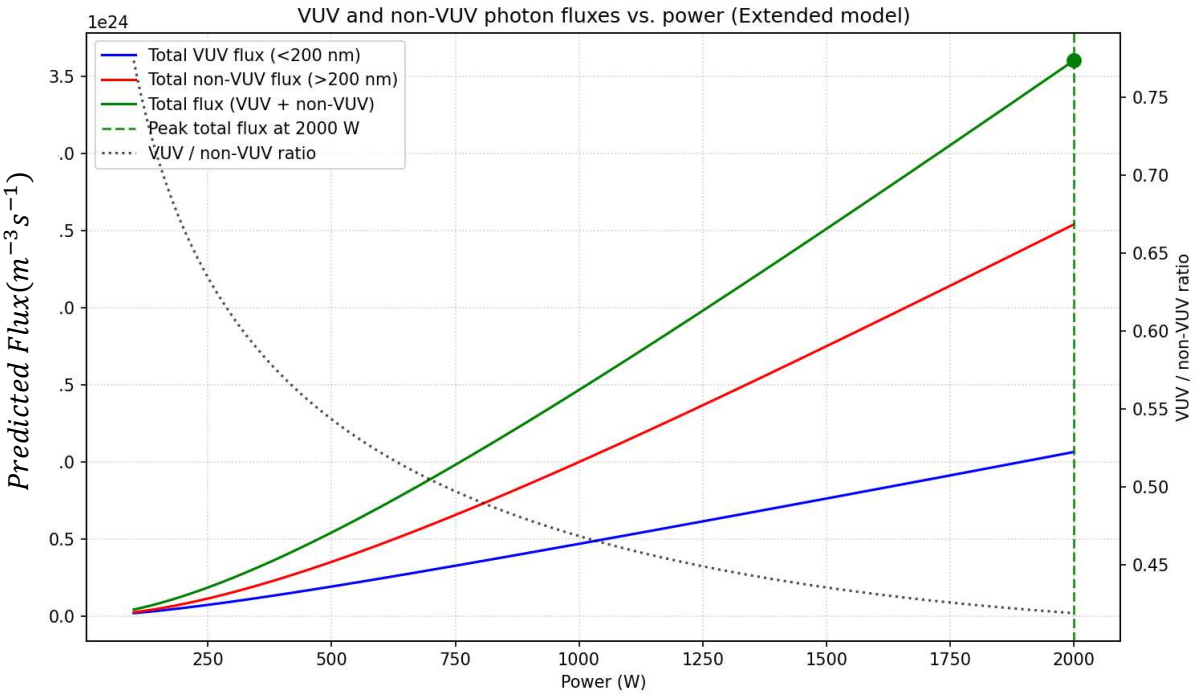


Fig 5.1.3.1. Graph showing the Total flux, total VUV flux, total non-VUV flux and VUV to non-VUV ratio for changing power

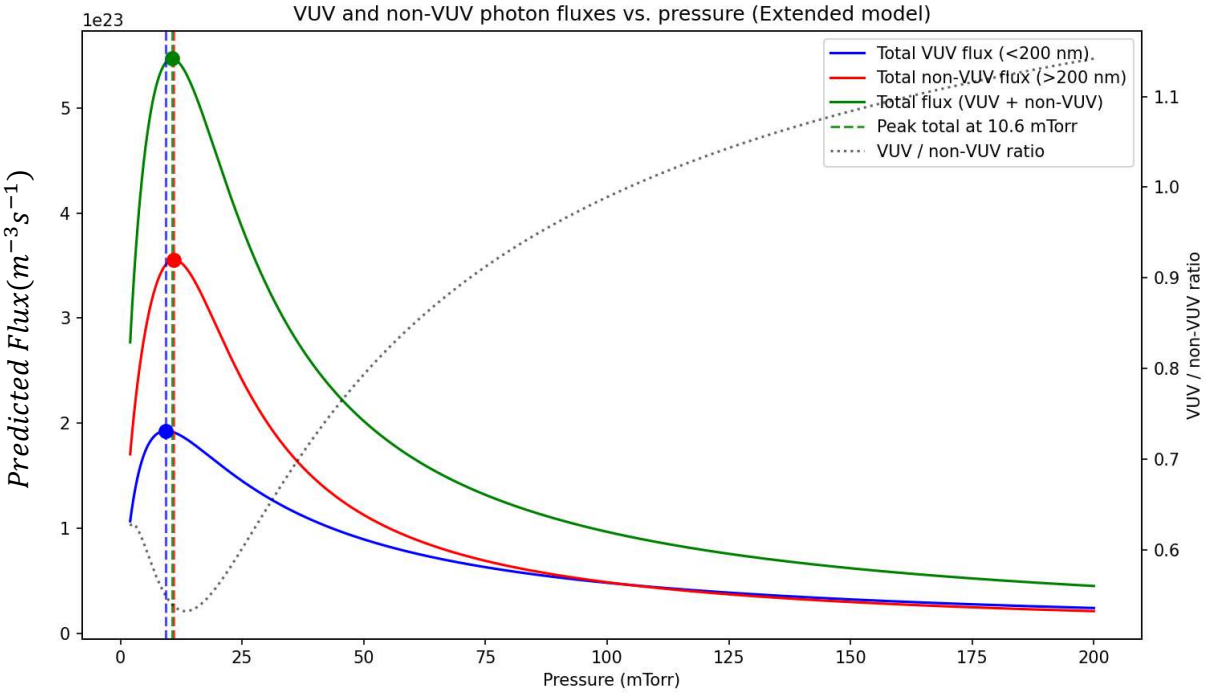


Fig 5.1.3.2. Graph showing the Total flux, total VUV flux, total non-VUV flux, VUV to non-VUV ratio and all of their peaks for changing pressure

5.2 Part B (Conjoined power and pressure variation)

5.2.1 2D Initial data acquisition

The same model parameters that were used in part A of this project remain the same for part B.

50 data points were taken, with 5 varying powers ranging from 200W to 2000W and 10 varying pressures ranging from 5mTorr to 200mTorr. The graph shown in figure 5.2.1.1 shows the how the resulting total photon emission rate changes with both power and pressure.

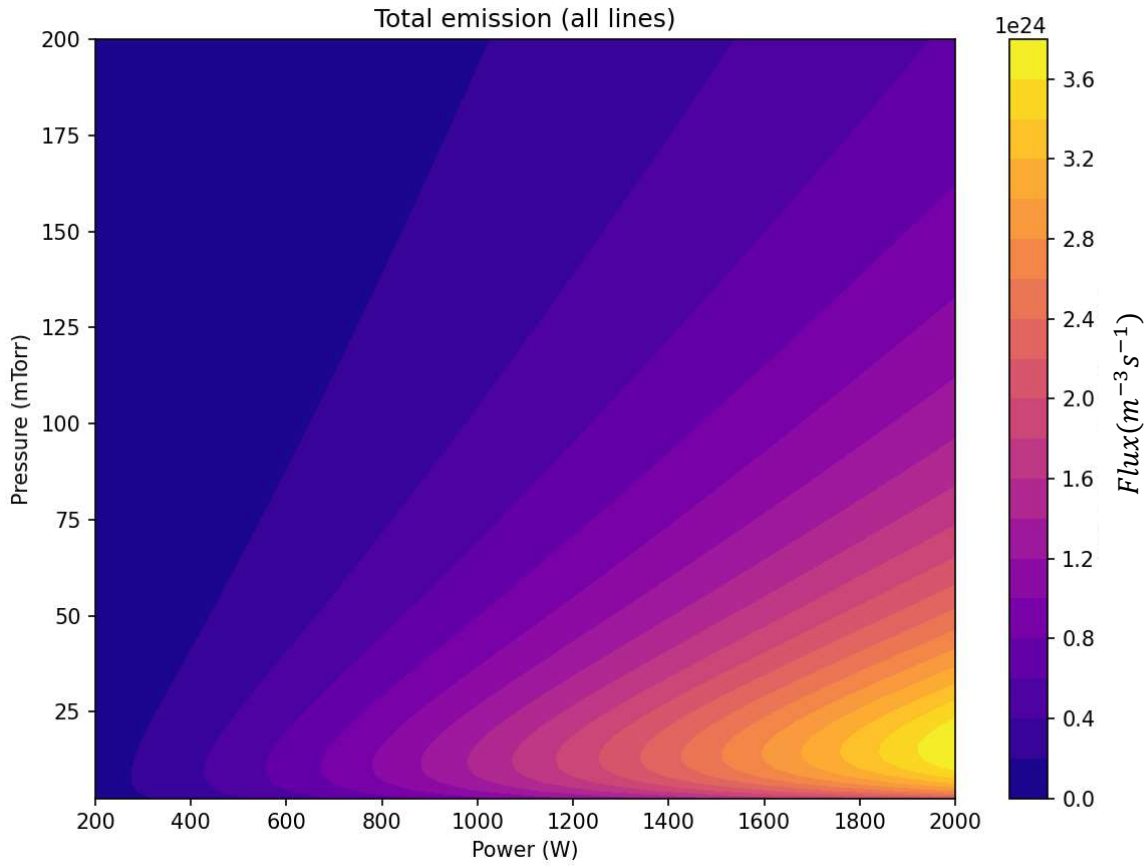


Fig 5.2.1.1. Graph showing the total emission rate for all emission lines in a 2D parameter space of pressure and power

5.2.2 2D Regression model implementation

Three models were then utilized to fit the 2D data:

- Combined 2D extended model: $I(p,P) = \frac{Ap^{\alpha}P^{\beta}}{1+Bp+CP+Dp^2+EP^2}$
- 2D exponential model: $I(p,P) = A \cdot p^{\alpha} \cdot P^{\beta} \cdot e^{-Bp-CP}$
- 2D extended model with interaction term: $I(p,P) = \frac{Ap^{\alpha}P^{\beta}}{1+Bp+CP+Dp^2+EP^2+FpP}$

The minimum R^2 value across all oxygen emission lines for each model can be found in figure 5.2.2.1. The resulting R^2 values from the models clearly show that the 2D extended model with

an interaction term models the data in this range the best. However, additional verification was still performed with all three models with a test data point of 1800W, 180mTorr. As this data point is not included in any of the training data points, it should show how well each model is interpreting the data.

Model	Minimum R^2
Combined 2D extended model	0.983519
2D exponential model	0.961424
2D extended model with interaction term	0.995926

Fig 5.2.2.1. Table showing the minimum R^2 across all emission lines for each of the 2D models

The interpolation data point again validated that the extended model with an interaction term modelled the photon flux data the best, with the largest error in photon flux emission prediction across all lines being 9.8%, followed by 13.6% for the extended model and 25.3% for the exponential model However, all models completely fail to predict the value of the O_5P -> O_5s emission line, with notably the extended model with an interaction term having an error of ~35%. To amend this, and with the knowledge that out of these three models, the interaction model is performing the best, a log space version of the model was implemented to attempt to amend these errors, as one of the possible causes for such errors could be the large values of the data being handled, and the large variation in the photon flux values across the given power and pressure ranges.

This results obtained from the new interaction model in log space can be seen in figure 5.2.2.2, where a worst case error of 9.3% and best case error of 1.13% occur. Whilst this model is not perfect and further improvements could be implemented to obtain higher degrees of accuracy, this model still simulates the changing of photon flux with varying power and pressure to a desired degree for the scope of this project, and allow for general predictions on what combinations of parameters will yield differing VUV, non-VUV and total flux.

Emission line	Logarithmic Interaction model prediction	Simulated value
r549: O_1s -> O_1d	4.6118e+18	4.30E+18
r550: O_5s -> O	1.0253e+21	1.05E+21
r551: O_5s -> O	3.3202e+20	3.41E+20
r552: O_3s -> O	1.6796e+23	1.64E+23
r553: O_3s -> O	9.9986e+22	9.75E+22
r554: O_3s -> O	3.3296e+22	3.25E+22
r555: O_5p -> O_5s	4.7419e+22	4.34E+22
r556: O_5p -> O_5s	4.7419e+22	4.34E+22
r557: O_5p -> O_5s	4.7419e+22	4.34E+22
r558: O_3p -> O_3s	5.0240e+22	4.96E+22
r559: O_3p -> O_3s	5.0240e+22	4.96E+22
r560: O_3p -> O_3s	5.0240e+22	4.96E+22

Fig 5.2.2.2. Table comparing the predicted values for each photon flux at 1800W and 180mTorr and the actual simulated value

Figure 5.2.2.3 shows the resulting graphs produced by this model showing how the total photon flux varies across the parameter space. Here the familiar spike in flux around 15 mTorr can be observed, as well as the general trend of increasing photon flux with increasing power, further reinforcing the results obtained in part A. It can also be observed that the graph shows very similar results to those obtained from the initial simulation data, shown in figure 5.2.1.1.

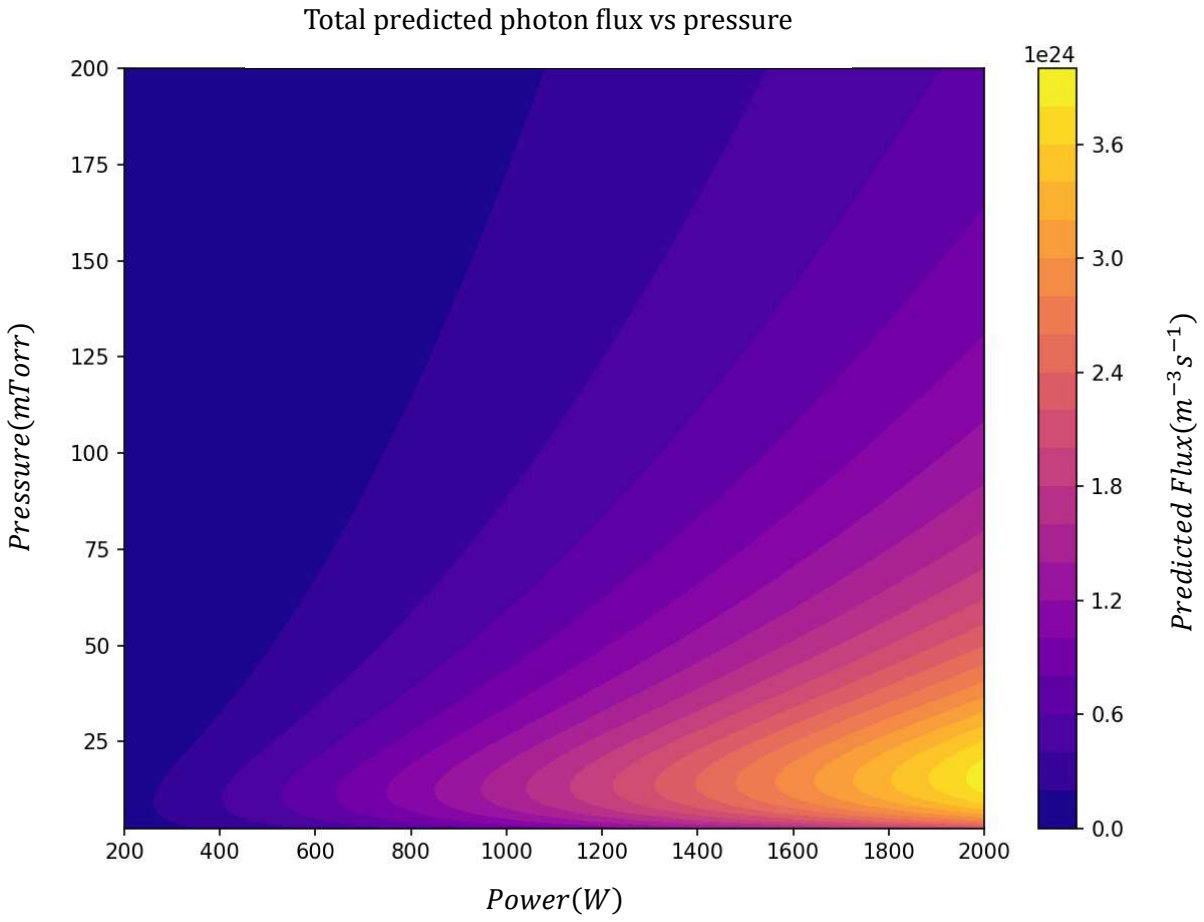


Fig 5.2.2.3. Graph showing how the predicted total photon flux varies over the 2D parameter space

5.2.3 2D Regression model analysis

Using this model, similar calculations to part A were performed to find the parameters that lead to: Peak flux, peak VUV flux, peak non-VUV flux, peak VUV to non-VUV ration and peak non-VUV to VUV ratio.

Figure 5.2.3.1 shows the resulting pressure and power combinations that correspond to each of these peaks. Here it can be observed that the maximum and minimum VUV to non-VUV ratios that can be obtained when varying both pressure and power in tandem are much more extreme than those obtained when only varying a single parameter. Figure 5.2.3.2 shows a graph showing how the VUV to non VUV ratio varies over the given parameter ranges.

Peak locations in the 2D parameter space			
Peak data	Pressure (mTorr)	Power (W)	Peak value
Peak total flux	15.24	2000.0	1.1397e+24 $m^{-}3s^{-1}$
Peak VUV flux	15.91	2000.0	2.7628e+24 $m^{-}3s^{-1}$
Peak non-VUV flux	15.91	2000.0	3.9018e+24 $m^{-}3s^{-1}$
Max VUV / non-VUV ratio	200.00	100.0	1.51
Min non-VUV / VUV ratio	2.00	2000.0	0.36

Fig 5.2.3.1. Table showing the peak values for total, VUV and non-VUV flux and maximum and minimum VUV to non-VUV ratios, along with the pressure and power values they occur at

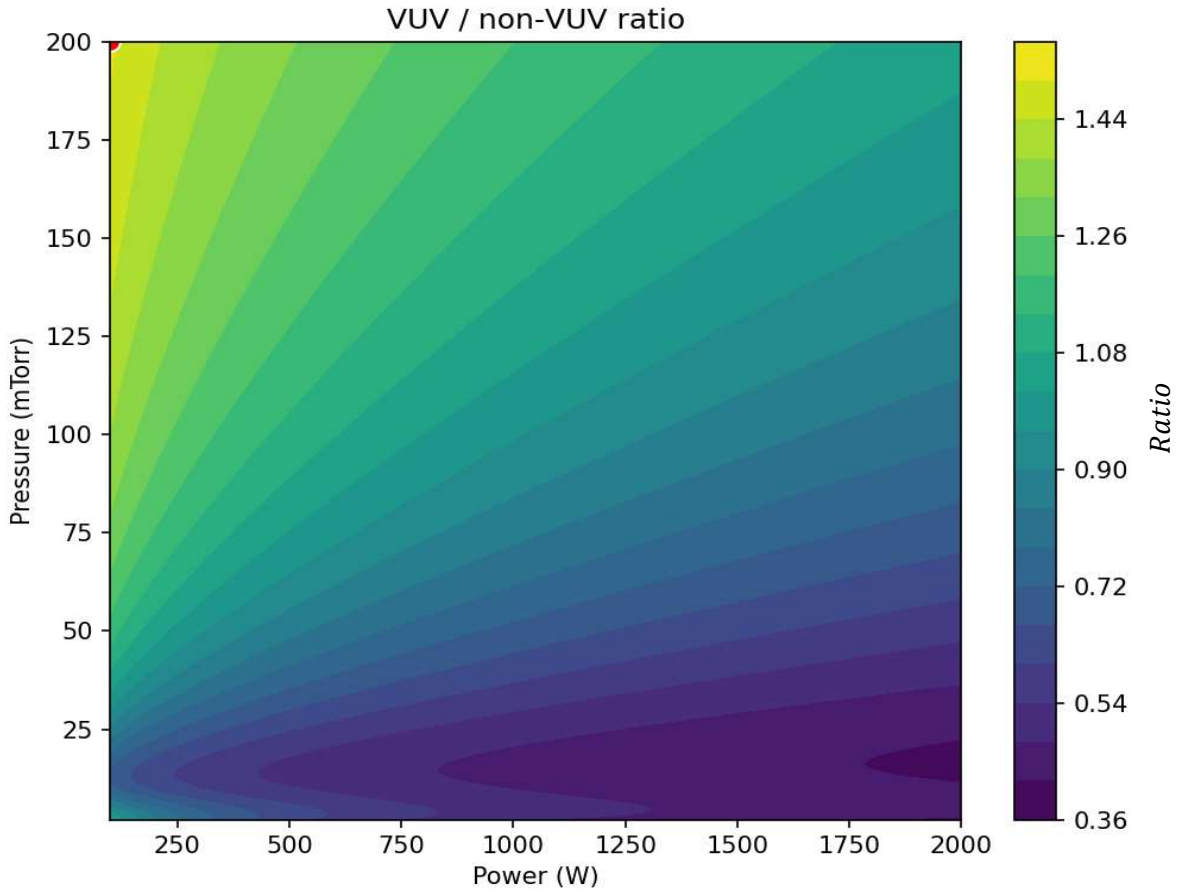


Fig 5.2.3.2. Graph showing how the VUV to non-VUV ratio varies over the given parameter ranges

6. Results analysis and discussion

6.1 Results analysis

The observed emission trends are consistent with oxygen-plasma emission physics described in the literature. In the Ar/O_2 ICP study by Engelbrecht et al., 2024, the dominant VUV features

were the 130 nm and 135 nm oxygen lines from the O($3s$) states, and the authors showed that the main production route for the 130 nm emission is a multi-step process involving excitation into O($3P$) followed by radiative decay through O($3S$). This provides a physically meaningful basis for interpreting the strong sensitivity of the simulated VUV output to both power and pressure.

At fixed pressure, increasing power raised the absolute photon flux for all lines, but with diminishing returns at the upper end of the range, which is captured well by the extended saturation fit reported in the Results section. This behaviour is consistent with the results of Engelbrecht et al. (2024), who found that while higher absorbed power increases the electron energy budget, the resulting photon emission becomes limited at high power by reaction-rate saturation, wall losses, and energy redistribution into competing inelastic channels. The falling VUV-to-non-VUV ratio with power therefore suggests that higher power increases overall excitation more rapidly than it preserves spectral selectivity toward VUV.

The pressure dependence is more informative from a kinetic standpoint. The existence of a clear pressure optimum for most emission lines is consistent with the oxygen-VUV paper, which reports that VUV emission peaks at intermediate pressure because electron density and O-atom-related production pathways are maximised in that regime, while further increases in pressure enhance loss processes such as negative-ion formation and collisional suppression. In this project, the same general balance appears in the form of a narrow pressure window where emission is strongest, followed by a rapid fall-off and eventual plateau at higher pressure.

6.2 Discussion

The ratio results show that pressure is the more effective single parameter to control for spectral balance, whereas power mainly controls absolute intensity. However, changing both of these parameters in tandem is deemed to be a much more effective way of controlling both of the spectral balance and absolute intensity simultaneously, with the VUV/non-VUV ratio reaching a maximum of 1.51 and a minimum of 0.36. While the operating point that gives the highest VUV selectivity may not coincide with the highest total photon flux, this is still useful from a processing perspective because it allows the plasma conditions to be chosen according to the desired outcome. For example, high power and intermediate pressure are favourable when strong overall photon flux is required, whereas higher pressure and lower power may be more appropriate when reducing VUV exposure is more important. This is important for PE-ALD as specifically VUV photon emissions can have a large effect on the process outcome, both positively and negatively, so controlling their emissions is desired.

These findings also show that regression-based modelling is a practical and efficient way of describing plasma emission behaviour across changing operating conditions; both the one-dimensional and two-dimensional regression models reproduced the simulation data well using relatively simple functional forms. This suggests that expanding the method to include additional plasma parameters would be a viable route to faster plasma optimisation, especially where trial-and-error tuning would otherwise be time-consuming. Further improvement would likely come from adding more datapoints and adopting a more physically informed model, but even in its current form the approach demonstrates that regression fits can provide a reliable and computationally efficient framework for predicting and tuning plasma emission properties.

7. Conclusion

Regression modelling of oxygen plasma photon emission has been shown to be a practical and efficient tool for predicting flux behaviour across power and pressure. Using a self-consistent global plasma model as a data source, an extended saturation function accurately captured the single-parameter trends ($R^2 > 0.99$), and a log-space interaction model reproduced the full two-dimensional power–pressure surface with worst-case errors below 10%. The analysis confirmed that pressure is the dominant control for the VUV-to-non-VUV spectral ratio, while power primarily scales absolute intensity, enabling operating points that range from a maximum VUV/non-VUV ratio of 1.51 (200 mTorr, 100 W) to a minimum of 0.36 (2 mTorr, 2000 W). These findings directly support the goal of mitigating VUV-induced substrate damage in PE-ALD by tuning process parameters. The results further demonstrate that regression models are a suitable method for accelerating plasma optimisation, making them a viable complement to more expensive physical simulations rather than experimental trial-and-error tuning.

8. Bibliography

Arts, K., Hamaguchi, S., Ito, T., Kazuhiro Karahashi, Harm, Mackus, A.J.M. and Kessels, E. (2022). Foundations of atomic-level plasma processing in nanoelectronics. *Plasma sources science & technology*, 31(10), pp.103002–103002. doi:https://doi.org/10.1088/1361-6595/ac95bc.

Engelbrecht, M.O., Jenderny, J., Hylla, H., Filla, D., Awakowicz, P., Ihor Korolov, Ridgers, C.P. and Gibson, A.R. (2024). Numerical investigation of vacuum ultra-violet emission in Ar/O_2 inductively coupled plasmas. *Plasma Sources Science and Technology*, 33(9), pp.095008–095008. doi:https://doi.org/10.1088/1361-6595/ad7059.

Profijt, H.B., Kudlacek, P., van and W. M. M. Kessels (2011). Ion and Photon Surface Interaction during Remote Plasma ALD of Metal Oxides. *Journal of The Electrochemical Society*, [online] 158(4), p.G88. doi:https://doi.org/10.1149/1.3552663.

Kessels, W.M.M., Mackus, A.J.M., Verheijen, M.A., van de Sanden, M.C.M. and Arts, K., 2007. *Remote plasma atomic layer deposition of metal oxides: fundamentals and applications*. Journal of Vacuum Science & Technology A, 25(4), pp.1023–1035. doi:https://doi.org/10.1116/1.2753846

SciPy (2019). *scipy.optimize.curve_fit — SciPy v1.3.1 Reference Guide*. [online] Scipy.org. Available at: https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.curve_fit.html.